PROJECT

Contents

1. Abstract

The main theorem in [Nak95] shows there is an action of the infinite dimensional Heisenberg algebra on the homology groups of the Hilbert scheme of points on a projective surface. We aim to explain the main construction and sketch the proof of the theorem.

2. Heisenberg algebra

This is mostly notes from video 8, except for Subsection 2.5, which is based on my meeting with Jethro. So far there's only stuff on Heisenberg.

2.1. **Lattice Heisenberg algebra.** Let's work out Emily's construction of the Heisenberg algebra in Video 8. Essentially she constructs the affine Lie algebra associated to a commutative Lie algebra (a lattice) and defines Heisenberg as the central extension of that. Though our construction of affine Lie algebras allowed for nonzero brackets in the original Lie algebra, in our definition of Heisenberg there's also no appearence of the bracket of the original Lie algebra: instead we take the underlying Lie algebra to be 1-dimensional! Perhaps that's why we call ours the "basic" Heisenberg algebra.

Let Γ be a lattice (i.e. a finitely generated torsion-free abelian group) equipped with a positive definite symmetric bilinear form $(\cdot,\cdot) : \Gamma \times \Gamma \to \mathbb{Z}$. E.g. $\Gamma = \sqrt{N}\mathbb{Z}$, $(\sqrt{N}n, \sqrt{N}m) = Nnm$.

Turn Γ into a commutative Lie algebra $\mathfrak{a} := \Gamma \otimes_{\mathbb{Z}} \mathbb{C}$ by putting zero bracket. Now consider the *loop algebra* $\mathfrak{a}[[t]]$, which is "spanned topologically" by $\alpha \otimes t^n$,

$\alpha \in \Gamma$ and $n \in \mathbb{Z}$. According to video 10 (6:40), the topology on $\mathfrak{g}[[t]]$ is defined via a basis of neighbourhoods of 0 of the form $t^n\mathfrak{a}[[t]]$. That is, the neighbourhoods get smaller and smaller as n gets bigger. We say that the loop algebra $\mathfrak{a}[[t]]$ is *spanned topologically* by $\alpha \otimes t^n$ if we first consider linear span of those elements and then the topological closure of that. (In our lectures, we introduced the same topology on the algebra $\mathbb{C}((t))$ when discussing Tate vector spaces.) Later in video 8 it is explained that we can replace this definition of loop algebra by ours, namely $L\mathfrak{a} = \mathfrak{a}[t, t^{-1}] = \mathfrak{a} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}]$, and study the representations without worrying about the topology.

Notice that $(\cdot, \cdot)$ extends to $\mathfrak{a}$.

Definition 2.2. The *Heisenberg Lie algebra* $\hat{\mathfrak{a}}_\Gamma$ associated to $(\Gamma, (\cdot, \cdot))$ is the central extension

$$0 \longrightarrow \mathbb{C}K \longrightarrow \hat{\mathfrak{a}}_L \longrightarrow \mathfrak{a}[[t]] \longrightarrow 0$$

with Lie bracket given by

- $[K, \hat{\mathfrak{a}}] = 0$, (this is why it is called *central* extension; K commutes with everything),
- and
$$[\alpha \otimes f(t), \beta \otimes g(t)]_{\hat{\mathfrak{a}}_L} := -(\alpha, \beta)(\mathrm{Res}_{t=0} f dg)K.$$

One checks that in fact this bracket satisfies the definition of Lie bracket, and that in monomials it acts by:

$$[\alpha \otimes t^m, \beta \otimes t^n] = -(\alpha, \beta)n\delta_{m,-n}K.$$

Compare this formula with Equation 6.2.1. It's identical except for a $(-1)^{m-1}$ factor. This is why we must introduce a variant of this Heisenberg algebra that has a superstructure, see Subsection 2.4.

From now on we denote $\alpha(n) := \alpha \otimes t^n$ for $\alpha \in \Gamma$.

2.3. **Universal enveloping algebra, Verma modules and Fock representation.** In this subsection we review basic Lie theory aiming to describe the Fock representation of the Heisenberg Lie algebra. Perhaps this can be entirely skipped and summarized in the claim that $F^0 = \mathbb{C}[x_1, x_2, \ldots]$ is a highest weight representation of $\hat{\mathfrak{a}}$.

We start with some remarks on the universal enveloping algebra of the Heisenberg Lie algebra. Recall that an explicit construction of the universal enveloping algebra of a Lie algebra is given by a quotient of the tensor algebra by an ideal generated by the commutation relations. Emily claims that the universal enveloping algebra of Heisenberg $U(\hat{\mathfrak{a}})$ contains finite linear combinations of finite products of possible *infinite* sums of $\alpha(n)$ and K. The "topological completion" of $U(\hat{\mathfrak{a}})$, denoted $\tilde{U}(\hat{\mathfrak{a}})$, is spanned by infinite linear combinations of finite products of the $\alpha(n)$ and K.

In this case there is a PBW theorem that says if $\alpha_1, \ldots, \alpha_r$ is a basis of Γ then $\tilde{U}(\hat{\mathfrak{a}})$ is "topologically spanned" by monomials of the form $\alpha_1(n_1^1)\alpha_1(n_1^2)\ldots\alpha_1(n_1^{k_1})\alpha_2(n_2^1)\ldots$, where $n_1^1 \leq n_1^2 \leq \ldots \leq n_1^{k_1}$. That is, we have monomials in infinitely many variables.

Like in the finite-dimensional case, any (continuous) $\hat{\mathfrak{a}}$-module is equivalent to a (continuous) $\tilde{U}(\hat{\mathfrak{a}})$-module. (Cf. [Kac10, Corollary 22.1]: any representation $\pi : \mathfrak{g} \to \mathrm{End}(V)$ of extends uniquely to a homomorphism of associative algebras $U(\mathfrak{g}) \to \mathrm{End}(V)$.)

Next consider the following characterization of irreducible $\hat{\mathfrak{a}}$-modules. Let V be an irreducible $\hat{\mathfrak{a}}$-module. Suppose that $K \in \mathrm{End}(V)$ has an eigenvalue (this will happen if we impose certain conditions on V like countable dimension, see Dixmier's lemma). Since K is central in $\hat{\mathfrak{a}}$, the endomorphism K is a morphism of $\hat{\mathfrak{a}}$-modules (indeed: a morphism of representations is just a $\hat{\mathfrak{a}}$-linear map).

Now choose an eigenvector v_0 with eigenvalue κ. Then the map $K - \kappa\mathrm{id}$ is also a morphism of $\hat{\mathfrak{a}}$-modules with nonzero kernel ($v_0 \in \mathrm{Ker}$). By Schur's Lemma, which says that any map of irreducible modules is either zero or an isomorphism, this map must be zero.

In conclusion, in any irreducible $\hat{\mathfrak{a}}$-module, K acts by multiplication by some complex number κ which we call the *level*. Often we just study representations of level κ (not worrying any more about the restrictions of V and barely asking that K acts by multiplication by κ).

Since $\hat{\mathfrak{a}}$-modules are equivalent to $\tilde{U}(\hat{\mathfrak{a}})$-modules, it turns out that $\hat{\mathfrak{a}}$-modules of level κ are equivalent to modules over the quotient $\tilde{\mathcal{H}}$ of the universal enveloping algebra by the ideal generated by $K - \kappa\mathrm{Id}$. Furthermore, any choice of $\kappa \neq 0$ gives an equivalent representation theory. Indeed, rescaling $\alpha(n) \mapsto \kappa^{1/2}\alpha(n)$ gives an isomorphism of representations. Notice this is not the case for any affine Lie algebra.

Now we turn to the Fock representation. In view of the construction so far we might as well consider level-1 representations, that is, K acts as the identity (as we did in our course from the very beginning!).

First we observe that it's not possible to have a 1-dimensional representation of $\tilde{\mathcal{H}}$ since for $\alpha \in \Gamma \setminus \{0\}$, $n \neq 0$ and $v \neq 0$ we would have $\alpha(n)v = \lambda v$ and $\alpha(-n)v = \mu v$, implying that

$$0 = \lambda\mu v - \mu\lambda v = [\alpha(n), \alpha(-n)v] = n\delta_{n,-n}K = nK,$$

which is not possible.

However, we can restrict ourselves to the subalgebra $\tilde{\mathcal{H}}_+ = \{\alpha(n) | n \geq 0, \alpha \in \Gamma\}$ and we no longer have this problem, so that in fact we have a trivial representation of $\tilde{\mathcal{H}}_+$ on $\mathbb{C}$. Then we can "induce" this representation to $\tilde{\mathcal{H}}$ and obtain the *Fock representation*

$$\pi := \mathrm{Ind}_{\tilde{\mathcal{H}}_+}^{\tilde{\mathcal{H}}} \mathbb{C} = \tilde{\mathcal{H}} \otimes_{\tilde{\mathcal{H}}_+} \mathbb{C}.$$

Let's quickly recall some facts about Verma modules following Video 10 from 31:00 to 39:04. See also [Kac10, Lecture 24].

Let $\mathfrak{g}$ be a semi-simple Lie algebra. Recall that for any Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ and subset of positive roots $\Delta_+ \subset \mathfrak{h}^*$ there's a triangular decomposition

$$\mathfrak{g} = \underbrace{\mathfrak{n}_-}_{\alpha(-n)} \oplus \underbrace{\mathfrak{h}}_{K} \oplus \underbrace{\mathfrak{n}_+}_{\alpha(n)}$$

where $\mathfrak{n}_\pm = \bigoplus_{\gamma\in\Delta_+} \mathfrak{g}_{\pm\gamma}$. The underbraces show how this is in analogy with the case of the Heisenberg algebra.

Now fix a character $\Lambda \in \mathfrak{h}^*$. Let $\mathfrak{n}_+ \curvearrowright \mathbb{C}$ be trivial and define an action $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+ \curvearrowright \mathbb{C}$ using Λ. We can extend to all of $\mathfrak{g}$ to obtain the *Verma module* $M(\Lambda)$ of highest weight Λ.

It follows from PBW theorem that $U(\mathfrak{g}) \cong U(\mathfrak{n}_-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}_+)$. In turn, this implies that $M(\Lambda) \cong U(\mathfrak{n}_-)$ as a vector space (see [Kac10, Lecture 24, Proposition

1.1(iii)']), which is isomorphic to a polynomial ring $\mathbb{C}[\alpha^- i]$, where $\alpha^- i$ is a basis of $\mathfrak{n}_-$.

In our case, this description allows us to conclude that in fact $\pi \cong \mathbb{C}[\alpha_i(n)]$ where $\alpha_i \in \Gamma$ is a basis of $\Gamma \otimes_{\mathbb{Z}} \mathbb{C}$ and $n < 0$. Further, the action of $\hat{\mathfrak{a}}$ is determined, and it is non other than the action of $H = \mathbb{C}[x_1, x_2, \ldots]$ we used in class. Switching to our more familiar notation, the action is given by

$$h_n \mapsto \begin{cases} n\frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0 \\ 0 & \text{if } n = 0 \end{cases}, \qquad K \mapsto \mathrm{Id}.$$

One checks that this is a highest-weight module with highest weight vector 1. We call the vector $1 \in \pi = H$ the *vacuum state*. (Here's π is Emily's notation and H is our notation.) The operators $\alpha(n)$ for $n < 0$ are called *creation operators*: we think that such an operator maps the vacuum (no particles) to the state $\alpha(n)$. For $n \geq 0$ we call $\alpha(n)$ *annihilation operators*: applying any such operator to a monomial enough times we can make it zero. We may write

$$\pi \cong \underbrace{\mathbb{C}}_{\text{vacuum}} \oplus \underbrace{\mathfrak{n}_-}_{\substack{\text{1-particle}\\ \text{states}}} \oplus \underbrace{\mathfrak{n}_- \otimes \mathfrak{n}_-}_{\substack{\text{2-particle}\\ \text{states}}} \oplus \ldots.$$

Introduce the *energy* as Δ... this gives the character formula

$$\chi = \prod_{m=0}^{\infty} \frac{1}{(1-q^m)^n}$$

for the Heisenberg algebra of rank n. In fact, for the rank 1 case we can easily see (after simply writing out each of the factors of the product and counting the number of appearances of each term) that

$$\prod_{m=0}^{\infty} \frac{1}{1-q^m} = \sum_{\Delta=0}^{\infty} p(\Delta) q^{\Delta}$$

where p is the number of integer partitions of Δ.

2.4. **Heisenberg superalgebra.** Nakajima's theorem shows that there exists representation of the Heisenberg *super*algebra on the homology of the Hilbert scheme of points on a surface. We can tell that it is indeed the super version of Heisenberg algebra (and not the usual Heisenberg algebra) by the $(-1)^{i-1}$ factor in Equation 6.2.1. In this section we introduce the Heisenberg superalgebra following Video 10. See also Video 8, 42:00 for a quick super remark and from 1:32:40 until the end.

Start by replacing Γ (or $\Gamma \otimes_{\mathbb{Z}} \mathbb{C}$) and $(-,-)$ with a super vector space $V = V_0 \oplus V_1$ and a super bilinear form, namely $\langle -,- \rangle : V \times V \to \mathbb{C}$ satisfying $\langle v, w \rangle = (-1)^{\deg v \deg w} \langle w, v \rangle$ for all v, w homogeneous.

The next step is pass to the loop algebra. That is, we have to bring in t. Here Josh makes the remark that Nakajima does not the use topological Lie algebra approach for the definition of the loop algebra. That is, he goes for the algebraic version, which is basically as in our lectures: the loop algebra is $V \otimes \mathbb{C}[t^{\pm 1}]$, but with one caveat: we don't include t^0. Let

$$W = V \otimes t\mathbb{C}[t] \oplus V \otimes t^{-1}\mathbb{C}[t^{-1}]$$

and put a bilinear form extending the one on V, namely, a bilinear form such that

$$\left\langle v\otimes t^i, w\otimes t^j\right\rangle = i\delta_{i,-j}\left\langle v,w\right\rangle.$$

And then, instead of using central extension, we do the following. The *Heisenberg superalgebra* associated to the super vector space V is

$$\hat{\mathfrak{h}}_V = \frac{\text{free Lie super algebra of } W}{[v,w]-\langle v,w\rangle 1}.$$

Notice that if we put a trivial super structure on V, namely if $V=V_0$, then we almost recover the old Heisenberg. The difference is that we do not have K and we do not have $v\otimes t^0$.

Finally, the Fock space (highest weight representation) associated to this algebra is

$$\pi = S^*(V\otimes t^{-1}\mathbb{C}[t^{-1}]), \qquad \alpha_i(n)$$

where $S^*(U)=T(U)/(a\otimes b-(-1)^{\deg a \deg b} b\otimes a)$ is the *super symmetric product* of U. This is an irreducible representation of $\hat{\mathfrak{h}}_V$.

Unfortunately it looks like there's something missing: Equation 6.2.1! I want to see here why the bracket of $\hat{\mathfrak{h}}_V$ is given exactly by that equation.

2.5. **Meeting.** To avoid mixing up everything too much, I put notes from my meeting with Jethro in this separate subsection.

Recall our definition of the "basic" Heisenberg Lie algebra: $\hat{\mathfrak{a}}=\bigoplus_{n\in\mathbb{Z}}\mathbb{C}h_n\bigoplus\mathbb{C}_K$ with bracket given by $[h_m,h_n]=m\delta_{m,-n}K$ and $[K,\hat{\mathfrak{a}}]=0$.

For a complex number $\mu\in\mathbb{C}$ define a highest weight module

$$F^\mu = U(\hat{\mathfrak{a}})\otimes_{U(\hat{\mathfrak{a}}_+)}\mathbb{C}|\mu\rangle.$$

Then $h_0|\mu\rangle=\mu|\mu\rangle$ ($\mu\in\mathbb{C}$). (The Heisenberg vertex algebra is F^0, i.e. $\mu=0$.)

It turns out that F^μ is irreducible. This is a special feature of $\hat{\mathfrak{a}}$, proved using Shapovalov form. For more complicated Lie algebras $\hat{\mathfrak{g}}$, Verma $\hat{\mathfrak{g}}$-modules can be non-irreducible.

Also, F^μ satisfies the universal property that if V is an $\hat{\mathfrak{a}}$-module with a vector $v\in V$ such that

$$h_0v=\mu v \qquad \text{and} \qquad h_{>0}v=0$$

then there exists a morphism of $\hat{\mathfrak{a}}$-modules

$$f: F^\mu \longrightarrow V$$
$$\text{such that } |\mu\rangle\longmapsto v.$$

Now, if $v\neq 0$, so $f(F^\mu)\neq 0$, then $f(F^\mu)\cong F^\mu/\operatorname{Ker}(f)$. But since F^μ is irreducible, $\operatorname{Ker}(f)=\{0\}$, i.e. $f:F^\mu\to V$ is an inclusion.

If, furthermore, V is generated by V, i.e. $U(\hat{\mathfrak{a}})\cdot v=V$, then $V=f(F^\mu)\simeq F^\mu$.

As corollary, any $\hat{\mathfrak{a}}$-module generated by a highest weight vector v (such that $h_0v=\mu v$) is isomorphic to F^μ.

Let's now focus on the "basic" Heisenberg Lie algebra, i.e. we fix $\mu=0$ from now on. Then $F^0\simeq\mathbb{C}[x_1,x_2,\ldots]$. Define the *energy* by

$$\Delta(x_1^{k_1}x_2^{k_2}\ldots x_s^{k_s})=\sum_j k_j.$$

(I think of Δ as a device to introduce a grading on F^0. In contrast, the natural grading on $\mathbb{C}[x_1, \ldots, x_n]$ is degree, so that the factors of the graded decomposition are generated by homogeneous monomials. Here, instead of the degree of a monomial we have it's energy.)

Then the character is

$$\sum_{\Delta=0}^{\infty} q^{\Delta} \dim(F^0_{\Delta}) = \prod_{m=1}^{\infty} \frac{1}{1-q^m}.$$

Consider the following similar construction. (This is Emily's construction!! Here we start with a 2-dimensional lattice, and later generalize to n-dimensional, and compute their characters.) Let $\mathfrak{h} = \mathbb{C}^2 = \mathbb{C}x + \mathbb{C}y$ and put the symmetric bilinear form given by $(x,x) = (y,y) = 1$ and (x,y) (i.e. with Gram matrix the identity). Equip it with a null bracket to make it an abelian Lie algebra. Then $\hat{\mathfrak{h}} = \mathfrak{h} \otimes \mathbb{C}[[t^{\pm}]] \oplus \mathbb{C}K$ has bracket $[at^m, bt^n] = \underbrace{[a,b]}_{=0} + m\delta_{m,-n}(a,b)K$. The Fock space/Verma module for $\hat{\mathfrak{h}}$ is $F^0 = \mathbb{C}[x_1, y_1, x_2, y_2, \ldots]$, and its character is

$$\sum q^{\Delta} \dim F^0_{\Delta} = \prod_{m=1}^{\infty} \frac{1}{(1-q^m)^2}.$$

In the quotient, one factor comes from the $x_1, x_2, \ldots$ and the other one from $y_1, y_2, \ldots$. Actually we can spell this out further. Notice that F^0 is spanned by monomials

$$x_1^{k_1} x_2^{k_2} \ldots x_2^{k_s} y_1^{\ell_1} y_2^{\ell_2} \ldots y_t^{\ell_t}.$$

Each of these monomials corresponds to exactly one term in the product

$$\begin{array}{ll} x \begin{cases} m=1 \\ m=2 \\ m=3 \end{cases} & \begin{array}{l} (1 + \underline{q} + q^2 + q^3 + \ldots)\times \\ (1 + q^2 + \underline{q^4} + q^6 + \ldots)\times \\ (\underline{1} + q^3 + q^6 + q^9 + \ldots)\times \end{array} \\ y \begin{cases} m=1 \\ m=2 \\ m=3 \end{cases} & \begin{array}{l} (\underline{1} + q + q^2 + q^3 + \ldots)\times \\ (1 + q^2 + q^4 + \underline{q^6} + \ldots)\times \\ (\underline{1} + q^3 + q^6 + q^9 + \ldots) \times \ldots \end{array} \end{array}$$

For instance, the monomial $x_1 x_2^2 y_2^3$ corresponds to the underlined term $(q)(q^4)(1)(1)(q^6)(1)\ldots$.

This construction generalizes for $U \cong \mathbb{C}^n$ with a symmetric non-degenerate bilinear form $(\cdot,\cdot) : U \times U \to \mathbb{C}$. There exists a basis $x^1, \ldots, x^n$ of U such that $(\cdot,\cdot)$ has Gram matrix the identity. Then the Fock module F^0 is spanned by monomials with are complicated to write and read, but the character is actually simple:

$$\prod_{m=1}^{\infty} \frac{1}{(1-q^m)^n}.$$

It's possible to keep track of the total number of x and y, as well as the total energy.

Recall from lectures that we found the character of the charged Fermions is

$$\chi_{F^{(m)}}(q) = \sum_{\Delta} \dim F^{(m)}_{\Delta} q^{\Delta} = q^{m/2} \prod_{k=1}^{\infty} \frac{1}{1-q^k}.$$

and that in fact the charge-c component in the (charge,energy)-decomposition of F^{ch} is in fact isomorphic to the Fock space $F^c = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|c\rangle$.

3. Variants of the Heisenberg algebra

After all the messy discussion in the previous section, all of it shall be gone and this shall be the only text surviving in the last version of the project.

It looks like Nakajima's representation of the Heisenberg algebra is not exactly a representation of the "basic" Heisenberg algebra we have been using in lectures. Here I introduce what seems to be the correct variant of the Heisenberg algebra following Video 8 from 1:35:49.

Recall our definition of the "basic" Heisenberg Lie algebra: $\hat{\mathfrak{a}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n \bigoplus \mathbb{C}K$ with bracket given by

$$[h_m, h_n] = m\delta_{m,-n}K, \qquad [K, \hat{\mathfrak{a}}] = 0. \tag{3.0.1}$$

In class we described it's Fock space representation $H = \mathbb{C}[x_1, x_2, \ldots]$ with

$$h_n \mapsto \begin{cases} n\frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0 \\ 0 & \text{if } n = 0 \end{cases}, \qquad K \mapsto \mathrm{Id}.$$

We introduced a grading in $H = \bigoplus_{\Delta=0}^{\infty} H_\Delta$ given by the *energy* of a monomial, defined by

$$\Delta(x_1^{k_1} x_2^{k_2} \ldots x_s^{k_s}) = \sum_j k_j.$$

This gives the character formula

$$\sum_{\Delta=0}^{\infty} q^\Delta \dim H_\Delta = \prod_{m=1}^{\infty} \frac{1}{1-q^m}. \tag{3.0.2}$$

In order to generalize this construction it's worth going over why this formula holds. Writing $x_m = at^{-m}$ we have

$$\begin{aligned} H &\cong \mathbb{C}[at^{-1}, at^{-2}, \ldots] \\ &\cong \mathrm{Sym}\left(\bigoplus_{m>0} \mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0} \mathrm{Sym}(\mathbb{C}at^{-m}) \\ &\cong \bigotimes_{m>0} (\mathbb{C} \oplus \mathbb{C}at^{-m} \oplus \mathbb{C}(at^{-m})^2 \oplus \ldots), \end{aligned} \tag{3.0.3}$$

since the symmetric power of a vector space is the polynomial ring in its generators, and the symmetric power of a direct sum is the tensor product of the symmetric powers of the factors.

It is not a coincidence that we can write the Fock module of this algebra as a symmetric power of a Lie subalgebra. This is due to the Poincaré-Birkoff-Witten theorem applied to the universal enveloping algebra of the $\mathfrak{n}_-$ term of the triangular decomposition of $\hat{\mathfrak{a}}$ as in the theory of finite dimensional Lie algebras (see [Kac10, Lecture 24]).

Putting a dummy variably q to every 1-dimensional factor and multiplying throughout, we obtain

$$\begin{aligned} m=1 &\qquad (1+q+q^2+q^3+\dots)\times \\ m=2 &\qquad (1+q^2+q^4+q^6+\dots)\times \\ m=3 &\qquad (1+q^3+q^6+q^9+\dots)\times\dots, \end{aligned}$$

so that indeed the character is given by Equation 3.0.2. (Regrouping according to the energy grading $H=\bigoplus_{\Delta=0}^{\infty} H_\Delta$ we see that this character must also equal $\sum_{\Delta=1}^{\infty} p(\Delta)q^\Delta$ where $p(\Delta)$ is the number of integer partitions of Δ.)

Another version of this construction we discussed in lectures (see [vE23, Section 3.1]) starts with any finite-dimensional vector space $\mathfrak{h}$ equipped with a symmetric bilinear form $(\cdot,\cdot)$ and define

$$\hat{\mathfrak{h}} = \mathfrak{h}\otimes\mathbb{C}[[t^{\pm1}]]\oplus\mathbb{C}K$$

with bracket

$$[at^m, bt^n] = m\delta_{m,-n}(a,b)K, \qquad [K,\hat{\mathfrak{h}}]=0, \tag{3.0.4}$$

which is just the affine Lie algebra of $\mathfrak{h}$ when regarded as a commutative Lie algebra.

A similar formula to Equation 3.0.3 is

$$\begin{aligned} H &\cong \mathrm{Sym}\left(\bigoplus_{m>0}\bigoplus_{a\in B}\mathbb{C}at^{-n}\right) \\ &\cong \bigotimes_{m>0}\bigotimes_{a\in B}\mathrm{Sym}(\mathbb{C}at^{-m}) \\ &\cong \bigotimes_{m>0}\bigotimes_{a\in B}\mathbb{C}[at^{-m}] \end{aligned} \tag{3.0.5}$$

where B is a basis of the underlying vector space $\mathfrak{h}$. If $\mathfrak{h}$ has dimension r, this has character

$$\prod_{m=0}^{\infty}\frac{1}{(1-q^m)^r}. \tag{3.0.6}$$

Taking this construction one step further, let's change $\mathfrak{h}$ for a finite-dimensional super vectorspace $\mathfrak{h}=\mathfrak{h}_0\oplus\mathfrak{h}_1$ with basis $B^{\mathrm{even}}\amalg B^{\mathrm{odd}}$. If we equip $\mathfrak{h}$ with a graded commutative pairing $\langle\cdot,\cdot\rangle$, we can apply the same construction as above to obtain the algebra

$$\hat{\mathfrak{h}} = (\mathfrak{h}_0[[t^{\pm1}]]\oplus\mathfrak{h}_1[[t^{\pm1}]])\oplus\mathbb{C}K \tag{3.0.7}$$

with Lie bracket

$$[at^m, bt^n] = \langle a,b\rangle\, m\delta_{m,-n}K, \qquad [K,\hat{\mathfrak{h}}]=0. \tag{3.0.8}$$

One can check that this is indeed a super Lie bracket (which essentially holds by the pairing being graded commutative and K being central), i.e. $\hat{\mathfrak{h}}$ is a Lie superalgebra.

Its Fock module is given by

$$\begin{aligned} H &\cong \mathrm{Sym}\left(\bigoplus_{m>0}\bigoplus_{a\in B}\mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0}\mathrm{Sym}\left(\bigoplus_{a\in B^{\mathrm{even}}}\mathbb{C}at^{-m}\right)\otimes\bigotimes_{m>0}\Lambda\left(\bigoplus_{a\in B^{\mathrm{odd}}}\mathbb{C}at^{-m}\right). \end{aligned}$$

Though it is not clear to me why the Fock representation becomes exterior power on the odd part — this must have to do with the Verma modules/universal enveloping algebras for super vector spaces.

Now recall that while the symmetric power of a vector space yields the polynomial algebra on the basis of the vector space, the exterior power yields the algebra generated by the wedge products of the basic vectors. In the case of a 1-dimensional vector space generated by the element of energy m, we get

$$\Lambda(\mathbb{C}at^{-m}) = \mathbb{C}\oplus\mathbb{C}at^{-m}$$

and thus the character contribution in this case is simply

$$1+q^m.$$

Multiplying throughout we obtain the character

$$\prod_{m=0}^{\infty}\frac{(1+q^m)^{\dim\mathfrak{h}^{\mathrm{odd}}}}{(1+q^m)^{\dim\mathfrak{h}^{\mathrm{even}}}}.$$

Let's introduce one last variant of this construction. This time we start from a graded vector space $\mathfrak{h}=\bigoplus_{m\in\mathbb{Z}}\mathfrak{h}_m$, which is graded as $\mathfrak{h}^{\mathrm{even}}=\bigoplus_{i\in\mathbb{Z}}\mathfrak{h}_{2i}$ and $\mathfrak{h}^{\mathrm{odd}}=\bigoplus_{i\in\mathbb{Z}}\mathfrak{h}_{2i+1}$. Suppose $\mathfrak{h}_i$ is finite-dimensional with basis B_i, and that $\mathfrak{h}$ is equipped with a graded commutative pairing $\langle\cdot,\cdot\rangle$. The definition of $\hat{\mathfrak{h}}$ as an algebra is given exactly as in Equation 3.0.7 with bracket given by Equation 3.0.8.

Now we obtain

$$\begin{aligned} H &\cong \mathrm{Sym}\left(\bigoplus_{m>0}\bigoplus_{\substack{a\in B_i\\ i\in\mathbb{Z}}}\mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0}\mathrm{Sym}\left(\bigoplus_{\substack{a\in B_{2i}\\ i\in\mathbb{Z}}}\mathbb{C}at^{-m}\right)\otimes\bigotimes_{m>0}\Lambda\left(\bigoplus_{\substack{a\in B_{2i+1}\\ i\in\mathbb{Z}}}\mathbb{C}at^{-m}\right). \end{aligned}$$

This gives the character (or Poincaré polynomial)

$$\sum_{\substack{\Delta\in\mathbb{Z}_{\geq}\\ i\in\mathbb{Z}}}\dim\mathfrak{h}_{i,\Delta}t^iq^{\Delta}=\prod_{m>0}\prod_j(1-(-1)^jt^jq^m)^{-(-1)^j\dim\mathfrak{h}_j}.$$

Finally fix $\mathfrak{h}=H_*(X)$, the singular homology ring of a complex projective surface X. The character of the corresponding Heisenberg algebra is

$$\prod_{m=1}^{\infty}\frac{(1+t^{2m-1}1^m)^{b_1(X)}(1+t^{2m+1}q^m)^{b_3(X)}}{(1-t^{2m-2}q^m)^{b_0(X)}(1-t^{2m}q^m)^{b_2(X)}(1-t^{2m+2}q^m)^{b_4(X)}} \tag{3.0.9}$$

It was proved by Göttsche that Equation 3.0.9 in fact equals

$$\sum_{n=0}^{\infty} q^n P_t(X^{[n]}),$$

where the Poincaré polynomials are defined by $P_t(Z) = \sum_{m\geq 0} b_m(Z)t^m$, $b_m(Z) = \mathrm{rank} H_m(Z)$ for any variety Z.

Since $\bigoplus_{n=0}^{\infty} H_*(X^{[n]})$ is the representation of the Heisenberg algebra we shall discuss in the main theorem, this shows that in fact it is a highest weight representation.

It remains to explain the strange factor $(-1)^{i-1}$ in the main equation Equation 6.2.1. The reason for including this term in the main theorem is explained in [Nak, Theorem 9.15], but I would expect to find it also in the bracket of our super variants of the Heisenberg algebra, namely in Equation 3.0.8...

4. Hilbert scheme of points

The discussion in this section will not be used in what remains of this project and may summarized by saying, as Anthony does in the very beginning of video 1, that the essential property of the Hilbert scheme is that a point (over $\mathbb{C}$) of $X^{[n]}$ is a 0-dimensional closed subscheme of X of length n.

Throughout X will be a projective complex surface. We start by introducing the Hilbert scheme following [Nak, Section 1.1]. We consider a contravariant funtor $\mathcal{H}ilb_X : (\mathrm{Schemes})^{\mathrm{op}} \to (\mathrm{Sets})$ which assigns to any scheme U the set of flat families of closed subschemes in X parametrized by U. Any such family is by definition a flat morphism

$$\begin{tikzcd} Z \arrow[d, "\pi"] & \subseteq X \times U \\ U, & \end{tikzcd}$$

so that the fibers of π are closed subschemes of X. The Hilbert polynomial at a fiber Z_u is defined by $P_u(m) = \chi(\mathcal{O}_{Z_u} \otimes \mathcal{O}_X(m))$. By flatness of π the Hilbert polynomial is constant along U [Har77, Part III, Theorem 9.9].

For a fixed polynomial P we define $\mathcal{H}ilb_X^P$ as the functor which maps U to the set of families over U with Hilbert polynomial P. A theorem by Grothendieck says $\mathcal{H}ilb_X^P$ is representable by a projective scheme Hilb_X^P. This means that there is a natural transformation $\mathrm{Hom}(-, \mathrm{Hilb}_X^P) \simeq \mathcal{H}ilb_X^P(-)$, that is, families over any scheme U are in bijection with morphisms $U \to \mathrm{Hilb}_X^P$. Moreover, as with any representable functor, we obtain a distinguished element associated to the identity morphism of Hilb_X^P. In our case, this distinguished element is a universal family $\mathcal{Z} \to \mathrm{Hilb}_X^P$ such that any other family $Z \to U$ is obtained as a pullback (the Hilbert functor acts on morphisms of schemes by pullbacks of families) of $\mathcal{Z}$. More exactly, the digram

$$\begin{tikzcd} \mathrm{id}_{\mathrm{Hilb}_X^P} \arrow[d, mapsto] & \mathrm{Hom}(\mathrm{Hilb}_X^P, \mathrm{Hilb}_X^P) \arrow[r] \arrow[d, "f^*"'] & \mathcal{H}ilb_X^P(\mathrm{Hilb}_X^P) \arrow[d, "f^*"] & \mathcal{Z} \arrow[d, mapsto] \\ f & \mathrm{Hom}(U, \mathrm{Hilb}_X^P) \arrow[r] & \mathcal{H}ilb_X^P(U) & f^*\mathcal{Z}. \end{tikzcd}$$

commutes.

This shows that studying families over the Hilbert scheme is enough to study any family of subschemes of our initial scheme X. Recall the fact that points x of any scheme X correspond to morphisms $\mathrm{Spec}(\kappa(x)) \to X$, where $\kappa(x)$ is the residue field of the point (see e.g. Stacks Project 01J9). Then for every point h of the Hilbert scheme we have a unique subscheme of X defined as the fiber

$$\begin{tikzcd} X_h \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{Z} \arrow[d] & \subset X \times \mathrm{Hilb}_X^P \\ \mathrm{Spec}(\kappa(h)) \arrow[r] & \mathrm{Hilb}_X^P. \end{tikzcd}$$

See Stacks Project 01K0 for the definition of fibre. We see that to make sure that the fibers are $\mathbb{C}$-schemes, then we must restricts ourselves to $\mathbb{C}$-points of the Hilbert scheme. This is just a technical remark specified in video 1.

Definition 4.1. Let P be a constant polynomial given by $P(m) = n$ for all $m \in \mathbb{Z}$. The *Hilbert scheme of n points in X* is $X^{[n]} := \mathrm{Hilb}_X^P$.

This condition forces the dimension of the fibers of any family over X to be zero. Indeed, the leading term of the Hilbert polynomial of a projective variety of degree d and dimension s is $(dm^s)/s!$, which can only be constant for $s = 0$ (see [HM98, Exercise 1.13]).

Further, any 0-dimensional subscheme of X is supported on finitely many points of X (this is consequence of X being Noetherian, see here: we couldn't have infinitely many points as it would give an infinite chain of prime ideals). Finally, the *length* of any such subscheme, defined by $\dim \Gamma(\mathcal{O}_Z)$ must be $\sum_{x \in \mathrm{supp} Z} \Gamma(\mathcal{O}_{Z_x}) = n$. (See Stacks Project 0AYT or Stack Exchange for the fact that if the support is 0-dimensional then $H^i(\mathcal{F}) = 0$ for $i > 0$ and thus the length of the scheme is what we just wrote.)

The reason why this scheme is called the Hilbert scheme of *points* is that any subscheme of X consisting of n distinct points is in Hilb_X^P. Indeed, if $Z = \{x_1, \ldots, x_n\} \subset X$, its structure sheaf is the direct sum of skyscraper sheaves

$$\mathcal{O}_Z = \bigoplus_{i=1}^{n} (\text{skyscraper sheaf at } x)$$

so that $\mathcal{O}_Z \otimes \mathcal{O}_X(m) = \mathcal{O}_Z$.

However, not every point in $X^{[n]}$ is given by a set of n distinct points, nor by a set of points with allowed repetitions. Indeed, the Hilbert scheme distinguishes the "tangent directions in which points collide". The important thing to note here is that there are elements in the Hilbert scheme given by ideals which do not correspond to sets of points (even with allowed repetition).

The following construction distinguishes the Hilbert scheme from the symmetric power $S^n X$ of the surface X.

The *symmetric power* of X is

$$S^n X = \underbrace{X \times \ldots \times X}_{n \text{ copies}} / S_n$$

where the symmetric group S_n acts by permutating copies. Thus, a point of S^nX is an S_n-orbit on the cartesian product $X \times \ldots \times X$, that is, an unordered n-tuple of points of X, with allowed repetition. In algebraic geometry these are called "algebraic cycles" and are denoted as formal sums $\sum_{x\in X} n_x[x]$ where $\sum_{x\in X} n_x = n$, with $n_x \in \mathbb{N}$ and $[x]$ are formal symbols.

Since elements of $X^{[n]}$ have length n as defined above, we have a map

$$\begin{aligned} \pi : X^{[n]} &\longrightarrow S^nX \\ Z &\longmapsto \underbrace{\sum_{x\in \mathrm{supp} Z} \dim \Gamma(\mathcal{O}_{Z,x})[x]}_{\text{cycle of } Z} \end{aligned}$$

called the *Hilbert-Chow morphism*.

As explained above, this map is an important tool used to distinguish the Hilbert scheme of a variety from its symmetric power. In some cases this is an isomorphism, but not always, as the following theorem suggests:

Theorem 4.2 (Fogarty)**.** *Suppose X is nonsingular and $\dim X = 2$, then the following holds.*

1. *$X^{[n]}$ is nonsingular and has dimension $2n$.*
2. *The Hilbert-Chow morphism is a resolution of singularities.*

5. Correspondences

Let's do a quick review of basic algebraic topology following [Hat00]. For every topological space we define a *singular n-simplex* to be a continuous map $\sigma : \Delta^n \to X$ where Δ^n is the simplex $\{(t_0, \ldots, t_n \in \mathbb{R}^{n+1} : \sum_i t_i = 1, t_i \geq 0 \forall i\}$. The free abelian group generated by the set of singular n-simplices in X is $C_n(X)$. Elements of $C_n(X)$ are formal sums $\sum_i n_i\sigma_i$ for $n_i \in \mathbb{Z}$. For every n there is a boundary map $\partial_n : C_n(X) \to C_{n-1}(X)$ such that $\partial_n\partial_{n+1} = 0$. A continuous map of topological spaces $f : X \to Y$ induces a homomorphism $f_\sharp : C_n(X) \to C_n(Y)$ by composition of a simplex $\sigma : \Delta^n \to X$ with f and extending linearly. This map satisfies $f_\sharp\partial = \partial f_\sharp$, thus giving a map in the category of chain complexes of groups. Thus we can apply the functor H_n to obtain an induced map on homology groups $f_* : H_n(X) \to H_n(Y)$.

Dual to the chain maps $f_\sharp$ are the cochain maps $f^\sharp : C^n(Y;G) \to C^n(X;G)$. The relation $f_\sharp\partial = \partial f_\sharp$ dualizes to $\delta f^\sharp = f^\sharp\delta$, so $f^\sharp$ induces homomorphisms $f^* : H^n(Y;G) \to H^n(X;G)$. Recall that the cohomology groups are constructed by considering the cochain groups $C_n^* := \mathrm{Hom}(C_n, G)$ and the coboundary map $\delta = \partial^* : C_{n-1}^* \to C_n^*$, this giving a cochain complex. While one could guess that $H^n(C;G)$ is isomorphic to $\mathrm{Hom}(H_n(C), G)$, this is overly optimistic and instead we have the universal coefficient theorem which relates them via the exact sequence

$$0 \longrightarrow \mathrm{Ext}(H_{n-1}(C), G) \longrightarrow H^n(C;G) \longrightarrow \mathrm{Hom}(H_n(C), G) \longrightarrow 0.$$

If $f : X \to Y$ is continuous, we have an induced map $f_* : H_i(X) \to H_i(Y)$. If a group G acts on X, then it acts on $H_\bullet(X)$ (this is a way producing a representation, it follows from the functoriality of homology).

Thinking of the graph of f as a subset of $X \times Y$, we can ask if a closed subset $Z \subset X \times Y$ (not necessarily the graph of a continuous map) will induce a map on homology $H_i(X) \to H_i(Y)$.

The natural idea is to produce a map using the projections

$$\begin{tikzcd} & Z \arrow[dl, "p_X"'] \arrow[dr, "p_Y"] & \\ X & & Y \end{tikzcd}$$

and essentially pulling back a class and then pushing it forward, i.e.

$$H_i(X) \xrightarrow{p_X^*} H_i(Z) \xrightarrow{p_{Y,*}} H_i(Y)$$

but it turns out that a functor cannot be covariant and contravariant at the same time, that is, the pushforward and pullback cannot be defined for all maps. So we shall give up functoriality and define the pushforward and pullback on some classes of maps instead.

Now we construct Borel-Moore homology, also known as locally finite homology. We always consider homology with $\mathbb{C}$ coefficients.

By *manifold* we mean a connected, oriented finite dimensional manifold over $\mathbb{R}$. The primary examples of manifolds will be smooth complex varieties. For example, V a smooth irreducible variety of dimension d over $\mathbb{C}$, which is a manifold of dimension $2d$, with the canonical orientation coming from the complex structure.

By *space* we mean a topological space that is a proper deformation retract of a closed neighbourhood of a manifold. For example, a variety V over $\mathbb{C}$. Also projective varieties are spaces seen as submanifolds of $\mathbb{P}^n$.

Let X be a space. Recall that simplicial homology is the homology of the singular chain complex

$$\cdots \longrightarrow C_i(X) \xrightarrow{\partial} C_{i-1}(X) \longrightarrow \cdots$$

where $C_i(X)$ is the set of finite linear combinations of i-simplices in X.

For Borel-Moore homology $H_\bullet^{BM}(X)$ we use $C_i^{lf}(X)$, which is the set of locally finite linear combinations of singular simplices. Here locally finite means that every point $x \in X$ is in a neighbourhood that intersects only finitely many simplices.

Borel-Moore homology is not homotopy invariant. To see this recall that the homology groups of $\mathbb{R}^2$ are $H_1(\mathbb{R}^2) = \mathbb{C}$ and $H_i(\mathbb{R}^2) = 0$ for $i \neq 1$. However, $H_2^{BM}(\mathbb{R}^2) = \mathbb{C}$ since a tiling of triangles is locally finite chain of 2 simplices, has no boundary (since the boundaries of all triangles cancel each other), and is clearly not the boundary of a 3-simplex; thus this tiling is a nonzero element of $H_2^{BM}(\mathbb{R}^2)$. Further $H_0^{BM}(\mathbb{R}^2) = 0$ since ray of 1-simplices is a locally finite chain whose boundary is a ponit; thus a point is a boundary and thus has trivial homology class.

Facts:

- If X is compact, $H_i^{BM}(X) \cong H_i(X)$.
- If X is not compact, $H_i^{BM}(X) \cong H_i(\hat{X}, \infty)$ where $\hat{X}$ is the one-point compactification of X and $H_i(\hat{X}, \infty)$ is the relative homology group.
- If the dimension of the manifold where X is embedded is m, there is a canonical isomorphism $H_i^{BM}(X) \cong H^{m-i}(M, M \setminus X)$. In other words, there is a perfect pairing
$$H_i^{BM}(X) \times H_{m-i}(M, M \setminus X) \to \mathbb{C}$$

given by “intersection” (where the quotes account for some transversal intersection condition that we would require before computing the intersection of two simplicial chains).

In the previous example of the tiling of $\mathbb{R}^2$ by triangles, we can take $X = M = \mathbb{R}^2$. To see how the intersection pairing works just intersect the 0-chain in singular homology with the tiling in Borel-Moore homology: since the point lies in exactly one triangle (after suitable perturbation) their intersection is 1. Then the pairing gives 1 between those two chains. Notice this wouldn't work singular homology: we need the infinite assumption on the 2-chain to make sure that the whole plane is tiled by triangles and we cannot move the point to a region where it is not contained in any triangle.

Key properties of Borel-Moore homology (see [Kam10, Section 4.2]):

(1) (Pushforward.) If $f : X \to Y$ is proper (e.g. inverse image of compact is compact) then $f_* : H_i^{BM}(X) \to H_i^{BM}(Y)$. This is functorial in proper maps, i.e. preserves compositions. This is because the inverse image of a locally finite intersection with a simplicial chain remains locally finite with respect to the preimage chain by properness.

(2) (Long exact sequence.) Let $U \subset X$ be open, $F = X \setminus U$, $u : U \hookrightarrow X$ and $f : F \hookrightarrow X$ inclusions. Then $u^* : H_i^{BM}(X) \to H_i^{BM}(U)$ fitting into a long exact sequence

$$\cdots \longrightarrow H_i^{BM}(F) \xrightarrow{f_*} H_i^{BM}(X) \xrightarrow{u^*} H_i^{BM}(U) \xrightarrow{\partial} H_{i-1}^{BM}(F) \longrightarrow \cdots$$

This can be constructed from the isomorphism above with singular cohomology.

(3) (Pullback.) If $f : X \to Y$ is a locally trivial fibration with fiber a manifold of dimension m (over $\mathbb{R}$), then $f^* : H_i^{BM}(Y) \to H_{i+m}^{BM}(X)$.

(4) (Intersection pairing.) If X and Y are both nicely embedded in the same manifold M,

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \to H_{i+j-m}^{BM}(X \cap Y).$$

The proof of this product is by using the isomorphism with relative cohomology, and then using usual cap product.

(5) (Fundamental classes.)
   - If V is a smooth irreducible variety of dimension d over $\mathbb{C}$ then let $[V]$ for the canonical generator of $H_{2d}^{BM}(V) \cong H^0(V) \cong \mathbb{C}$. (Where the last isomorphism is just evaluating the 0-cochain and evalating in the only point that exists, up to homology, since V is connected.) This is the *fundamental class* of V.
   - If V is an irreducible variety of dimension d over $\mathbb{C}$, let $[V] \in B_{2d}^{BM}(V)$ be the unique d such that $u^*[V] = [V_{\text{reg}}]$. That is, $[V]$ is the class corresponding to the pullback of u on the open set of regular points of V. The pullback in this degree is injective and thus this class is unique.
   - For V not necessarily irreducible of dimension d, then $H_{2d}^{BM}(V)$ has basis $[V_1], \ldots, [V_k]$ where V_i are the d-dimensional irreducible components of V. In this setting, the fundamental class is defined as the sum of the fundamental classes of those components, i.e. $[V] = [V_1] + \ldots + [V_k]$.

A special feature of BM homology is that we can define these fundamental classes. In particular, the fundamental class of the whole space, as in our example with $\mathbb{R}^2$, is a BM-class because we admit infinite chains.

Finally we arrive at correspondences.

Situation 5.1. Let X_1, X_2 be smooth irreducible varieties of dimensions d_1, d_2. Let Z be a d-dimensional closed subvariety of $X_1 \times X_2$ such that the projections p_1 and p_2 are proper.

The *correspondence* associated to $Z \subset X_1 \times X_2$ is the map

$$\begin{aligned} c_{[Z]} : H_i^{BM}(X_2) &\longrightarrow H_{i+2d-2d_2}^{BM}(X_1) \\ c &\longmapsto p_{1,*}(p_2^* c \cap [Z]) \end{aligned} \tag{5.1.1}$$

where p_1 and p_2 are the projections in the following diagram:

$$\begin{tikzcd} & Z \subset X_1 \times X_2 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 & & X_2. \end{tikzcd}$$

That is, it is given by pulling back a class by p_2, intersecting with the fundamental class of $[Z]$, and pushing forward by p_1.

Let us explain with more details why this map is well-defined. First we apply the pullback

$$H_i^{BM}(X) \xrightarrow{p_2^*} H_{i+2d_1}^{BM}(X_1 \times X_2),$$

which is valid by property (3) of BM cohomology since the projection onto the second factor is a locally trivial (in fact trivial) fibration with fibers of real dimension $2d_1$. Also by (3) we see that the degree of the target is $i + 2d_1$.

Then we take the intersection pairing with the fundamental class of Z, which lives in the $2d$-th BM homology of Z. According to (4) this should be a map

$$H_{i+2d_1}^{BM}(X_1 \times X_2) \xrightarrow{\cap_{X_1 \times X_2}[Z]} H_{i+d_1+2d-(2d_1+2d_2)}^{BM}(Z),$$

where the degree of the target is given by

$$\underbrace{i + 2d_1}_{\substack{\text{we are here} \\ \text{from previous step}}} + \underbrace{2d}_{\substack{\text{real dimension} \\ \text{of } Z}} - \underbrace{(2d_1 + 2d_2)}_{\substack{\text{real dimension} \\ \text{of ambient space}}},$$

which of course equals $i + 2d - 2d_2$.

Finally take the pushforward map of p_1 using property (1), since we assumed p_1 is proper. All in all, using pullback, intersection and pushforward we have constructed

$$H_i^{BM}(X) \xrightarrow{p_2^*} H_{i+2d_1}^{BM}(X_1 \times X_2) \xrightarrow{\cap_{X_1 \times X_2}[Z]} H_{i+2d-2d_2}(Z) \xrightarrow{(p_1)_*} H_{i+2d-2d_2}(X_1).$$

Morally we think of the first two steps as just doing pullback from X_2 to Z, but this can't be done so simply since we don't want to assume that p_2 is locally trivial (this won't be true in practice), and that's the only case when we get pullbacks. So that's why we pullback to a larger space where Z is contained and then take intersection.

Remark 5.2. (1) We can define correspondences more generally $c_\alpha : H_1^{BM}(X_2) \to H^{BM}_{i+2d-2d_2}(X_1)$ for any $\alpha \in H^{BM}_{2d}(Z)$, i.e. not only the fundamental class of a subvariety Z.

(2) If Z is the graph of a proper morphism $f : X_2 \to X_1$, then $c_{[Z]} = f_*$. So that in fact correspondences generalize the usual pushforward notion we have for BM homology.

(3) We may use Poincaré duality (either in its BM version $H_i^{BM}(M) \cong H^{m-i}(M)$ or the usual $H_i(M) \cong H^{m-i}(M)$ for any compact M) to identify a class in the tensor product with an endomorphism as we do in general for V, W vector spaces via the isomorphism $V \otimes W^* \cong \mathrm{Hom}(W, V)$, $v \otimes \varphi \mapsto (w \mapsto \varphi(w)v)$. This way, the fundamental class $[Z] \in \bigoplus_{i+j=\dim_{\mathbb{R}} Z} H_i(X_1) \otimes H_j^{BM}(X_2)$ immediately gives a map

$$\bigoplus_n H_*(X^{[n]}) \to \bigoplus_n H_*(X^{[n \mp i]})$$

which coincides with Equation 5.1.1.

Next we show how to compose correspondences.

Suppose we are given two correspondences as subvarieties $Z_{12} \subset X_1 \times X_2$ and $Z_{23} \subset X_2 \times X_3$. Their composition should be a class in the homology ring of $X_1 \times X_3$. We define the composition of the corresponding correspondences by

$$[Z_{12}] \circ [Z_{23}] := p_{13,*}(p_{12}^*[Z_{12}] \cap p_{23}^*[Z_{23}]) \tag{5.2.1}$$

according to the diagram

$$\begin{tikzcd}
 & X_1 \times X_2 \times X_3 \arrow[dl, "p_{12}"'] \arrow[dr, "p_{23}"] \arrow[dd, "p_{13}"'] & \\
X_1 \times X_2 & & X_2 \times X_3 \\
 & X_1 \times X_3 &
\end{tikzcd}$$

which is well defined since, in fact, the projection $p_{13}(p_{12}^{-1} Z_{12} \cap p_{23}^{-1} Z_{23}) \to X_3$ is proper (as required to apply the pullback according to property (3) of BM homology).

In fact, this operation can also be thought of via the pullback in the diagram

$$\begin{tikzcd}
 & & Z_{12} \times_{X_2} Z_{23} \arrow[dl] \arrow[dr] & & \\
 & Z_{12} \arrow[dl] \arrow[dr] & & Z_{23} \arrow[dl] \arrow[dr] & \\
X_1 & & X_2 & & X_3.
\end{tikzcd}$$

In fact, the composition of the correspondences in the bottom triangles is the correspondence of the outer triangle. Here we think of the fiber product as

$$\begin{aligned} Z_{12} \times_{X_2} Z_{23} &= \{(x_1, x_2, x_3) \in X_1 \times X_2 \times X_3 : (x_1, x_2) \in Z_{12}, (x_2, x_3) \in Z_{23}\} \\ &= p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}). \end{aligned}$$

We get

$$Z_{12} \circ Z_{23} = p_{13}(p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23})$$

so that

$$Z_{12} \circ Z_{23} = \{(x_1, x_3) \in X_1 \times X_3 : \exists x_2 \in X_2 \text{ s.t. } {\scriptstyle (x_1,x_2)\in Z_{12} \atop (x_2,x_3)\in Z_{23}}\}.$$

So that if Z_{12} and Z_{23} where the graphs of two functions, then $Z_{12} \circ Z_{23}$ would be the graph of their composition. Based on this observation, it would be nice if

$$c_{[Z_{12}]} \circ c_{[Z_{23}]} = c_{[Z_{12} \circ Z_{23}]},$$

which happens in the case of graphs, but this is false in general. Tracing through the definitions, we see it's one of these more general correspondences that is associated to a class that is not the fundamental thing of a single thing. So instead of taking the fundamental class of $Z_{12} \circ Z_{23}$, we get

$$c_{[Z_{12}]} \circ c_{[Z_{23}]} = c_{(p_{13})_*(p_{12}^*[Z_{12}] \cap_{X_1 \times X_2 \times X_3} p_{23}^*[Z_{23}]},$$

which is an element of $H^{BM}_{2\dim Z_{12} + 2\dim Z_{23} - 2\dim X_2}(Z_{12} \circ Z_{23})$ as one can check.

Example 5.3. (See the second part of video 9.) As an example of this construction, we may define an $\mathfrak{sl}_2$-action on $\bigoplus_{a,i} H_i^{BM}(T^*\mathrm{Gr}(a, N))$. where $\mathrm{Gr}(a, N)$ is the Grassmanian. Writing $\mathfrak{sl}_2 = \mathbb{C}\{e, h, f\}$, the idea is to construct the maps representing e and f as correspondences. The task is to define a subvariety Z of the product of the appropriate cotangent bundles, and prove that the associated correspondences satisfy the commutativity relations that they should. Further, this representation is irreducible since it happens have 1-dimensional weight spaces, which characterizes the only irreducible representation of $\mathfrak{sl}_2$.

6. Main construction

We want to produce a representation π of the Heisenberg (super)algebra $\mathfrak{h}$ on $V = H_*(X)$. The key idea is to consider some subvarieties of direct products and use their fundamental classes to produce operators on the homology via correspondences.

Let $n \geq i > 0$. Define $P[i]$ as a subset of $\coprod_n X^{[n-i]} \times X^{[n]}$ by

$$P[i] := \coprod_n \{(\mathcal{J}_1, \mathcal{J}_2, x) \in X^{[n-i]} \times X^{[n]} : \mathcal{J}_1 \supseteq \mathcal{J}_2, \mathrm{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\} \text{ for some } x \in X\}.$$

For $i < 0$, we swap $\mathcal{J}_1$ and $\mathcal{J}_2$ in this definition.

Notice that by the procedure in Section 5 the submanifold $P[i]$ already defines a correspondence by pullback, intersection with the fundamental class of $P[i]$ and then pushing forward. However, this is not quite what we will do: we will bring in two other classes. Let $\alpha \in H_*^{BM}(X)$, $\beta \in H_*(X)$ and $i > 0$. Define

$$\begin{aligned} P_\alpha[i] &= \varpi_*(\Pi_1^*\alpha \cap [P[i]]) \\ P_\beta[i] &= \varpi_*(\Pi_1^*\beta \cap [P[-i]]) \end{aligned}$$

according to the projections

$$\begin{tikzcd}[column sep=small] & \coprod_n X^{[n-i]} \times X^{[n]} \times X \arrow[dl, "\Pi_1"'] \arrow[dr, "\varpi_i"] & \\ X & & \coprod_n X^{[n-i]} \times X^{[n]}. \end{tikzcd}$$

I think the reason for this is that using just $P[i]$ as the correspondence could give non-proper projections, and the pullback may not be well-defined. Also, it's not clear why we have to choose α and β in different kinds of homology. Of course if we just assume that X is projective then BM homology coincides with usual homology.

In conclusion, the classes $P_\alpha[i]$ and $P_\beta[i]$ belong to $\prod_n H_*(X^{[n\mp i]}) \otimes H_*(X^{[n]})$. Here the $\mp$ means that if $i > 0$ we are substracting, and if $i < 0$ we are adding.

Now we use Poincaré duality (either in its BM version $H_i^{BM}(M) \cong H^{m-i}(M)$ or the usual $H_i(M) \cong H^{m-i}(M)$ one assuming M is compact) to identify a class in the tensor product with an endomorphism, as in $V \otimes W^* \cong \mathrm{Hom}(W, V)$, $v \otimes \varphi \mapsto (w \mapsto \varphi(w)v)$. This way $P_\alpha[i]$ and $P_\beta[-i]$ give maps

$$\bigoplus_n H_*(X^{[n]}) \to \bigoplus_n H_*(X^{[n\mp i]}).$$

It's worth noting that in practice we will use the construction in Section 5, specifically in Remark 5.2(1) for the classes $P_\alpha[i]$ and $P_\beta[i]$. Namely, for $i > 0$ will use the projections p_1 and p_2 in

$$\begin{tikzcd}[column sep=small] & X^{[n-i]} \times X^{[n]} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X^{[n-i]} & & X^{[n]}. \end{tikzcd}$$

and the formula

$$P_\alpha[i](c) = p_{1,*}(p_2^* c \cap \Pi_1^* \alpha). \tag{6.0.1}$$

The following proposition is used to compute the exact degrees in the domain and codomain of our correspondences:

Proposition 6.1. $\dim_{\mathbb{C}} P[i] \cap (X^{[n-i]} \times X^{[n]}) = 2n - i + 1$.

Proof. Suppose $i > 0$. By Theorem 4.2 we know that $\dim X^{[n-i]} = 2(n-i)$. Fix $\mathcal{J}_1 \in X^{[n-i]}$. We wish to find the dimension of the variety obtained by varying $\mathcal{J}_2 \in X^{[n]}$ satisfying $\mathcal{J}_1/\mathcal{J}_2 = \{x\}$ for some $x \in X$. The generic case is when $x \in X \setminus \mathrm{supp}(\mathcal{O}_X/\mathcal{J}_1)$ (this may be accepted under the explanation that being $\mathcal{O}_X/\mathcal{J}_1$ the structure sheaf of a variety, it has less dimension than that of X, i.e. it is generic that x is not in this support; a formal proof is provided in [Nak, Lemma 8.31] during the proof of the main theorem). In fact, the set of such $\mathcal{J}_2$ may be identified with $\pi^{-1}(S^i_{(i)}X)$, where π is the Hilbert-Chow morphism and $S^i_{(i)} = \{[i]x \in S^i(X) : x \in X\}$. It is shown in [Nak, Lemma 6.10] that it must have complex dimension $i + 1$. □

Now we are almost ready to formulate the main theorem. First notice that this is [Nak, Theorem 8.13] and **not** [Nak95, Theorem 3.1]. Now let's clarify what are the main objects involved.

The theorem is about a representation of the Heisenberg superalgebra introduced in Section 3 for the graded (and thus super) vector space $\mathfrak{h} = H_*(X)$. This means that we will construct a representation of the infinite-dimensional Heisenberg superalgebra $\hat{\mathfrak{h}}$ given by Equation 3.0.7 with bracket given Equation 3.0.8.

The pairing we use is the graded commutative pairing defined by $\langle \alpha, \beta \rangle = p_*(\alpha \cap \beta)$ where $p : X \to \mathrm{pt}$ is the unique morphism from X to the point pt. Notice that this pairing is indeed graded commutative since the intersection pairing $\cap$ in BM homology is introduced via the usual cup product in relative cohomology (see key property (4) of BM homology); and of course it is valued in $\mathbb{C}$ since $H_*(\mathrm{pt}) = \mathbb{C}$.

Theorem 6.2 (Nakajima, Grojnowski). *Suppose* $(deg\alpha)(deg\beta) \equiv 0(mod 2)$. *Then*

$$[P_\alpha[i], P_\beta[j]] = (-1)^{i-1} i \delta_{i,-j} \langle \alpha, \beta \rangle \, Id. \tag{6.2.1}$$

Notice the condition on the degrees of α and β. *If we ignore this*, then we immediately interpret the theorem as saying that this construction is a representation of $\hat{\mathfrak{h}}_V$ on $\bigoplus_n H_*)(X^{[n]})$. Another way to say this is: if $V = H_*(X)$ is concentrated in even degrees, then $\bigoplus_n H_*(X^{[n]})$ is a representation of $\hat{\mathfrak{h}}_V$. Further, this representation has highest weight vector the generator of $H_0(X^{[0]}) \cong \mathbb{Q}$.
Note. I still don't understand why the $(-1)^{i-1}$ appeared: this was not in the definition of the Heisenberg superalgebra as a quotient of the Lie algebra. I think that is due to the fact that that definition is done only for a super vector space, and not only a $\mathbb{Z}$-graded vector whose whose super structure is just the even/odd structure inherited from the $\mathbb{Z}$-grading.

It is not immediate that this representation is irreducible, but that is in fact true.

Theorem 6.3 (Nakajima). *This representation is irreducible by considering graded dimensions, which we will not discuss.*

7. Worked example

In this section we do a worked example following [Nak] and Video 10 to illustrate the main idea of the proof.

Fix $x \in X$ and let $\alpha = [X]$ be the BM class of X, and $\beta = [x]$ the class of the point x. Take a set of distinct n points $x_1, \dots, x_n \in X \setminus \{x\}$ none of which is x. Identify them with the corresponding ideal $\mathcal{J}$. We are going to interchangeably refer to those points as points and as the sheaf $\mathcal{J}$.

Let's calculate

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}].$$

Looking back at Equation 6.2.1, it's easy to see that we should obtain just Id.

Let's compute first $P_\beta[-1][\mathcal{J}]$. For this consider

$$\begin{tikzcd} & X^{[n+1]} \times X^{[n]} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X^{[n+1]} & & X^{[n]}. \end{tikzcd}$$

By definition, $P_{[x]}[-1]$ is given by: pull back $\mathcal{J}$ via p_2, intersect with the class $\varpi_*(\Pi^*\alpha \cap [P[-1]]) = P_{[x]}[-1]$ and pushforward down to $X^{[n+1]}$. Morally this means that we pushforward the cycle corresponding to

$$\{(\mathcal{J}, \mathcal{J}', x) : \mathcal{J} \subseteq \mathcal{J}', \mathrm{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}\} \subset X^{[n]} \times X^{[n+1]} \times X.$$

The pushforward recovers the $X^{[n+1]}$ component, that is, the schemes $\mathcal{J}'$ in $X^{[n+1]}$ satisfying $\mathrm{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}$. Since x is not a point in $\mathcal{J}$, the only possible choice is that $\mathcal{J}' = \{x\} \cup \mathcal{J}$, that is,

$$P_{[x]}[-1][\mathcal{J}] = [\{x\} \cup \mathcal{J}].$$

On the other hand, the correspondence $P_{[X]}[1]$ starts with an ideal $\mathcal{J}' \in X^{[n+1]}$ intersects it with

$$\{(\mathcal{J}, \mathcal{J}', x) : \mathcal{J} \subseteq \mathcal{J}', \mathrm{supp}(\mathcal{J}'/\mathcal{J}) = \{y\}, \text{ for some } y \in X\}$$

and projects to $X^{[n]}$. We obtain $\sum_{y \in \mathcal{J}'} [\mathcal{J}' - \{y\}]$. Then it becomes clear that

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}]$$

(see [Nak, Figure 8.1]).

The idea is very simple: it is not the same to add the point x and then remove any point, than removing any point and then adding the point x. Indeed, in the latter case we are obliged to have x, while in the former we may or may not. Thus the operators do not commute.

Notice that this is a very particular situation by the choice of $i = 1$ and $j = -1$. For different choices of i and j we obtain schemes such as double points which cannot be thought of barely as sets of points —and pictures do not accurately represent the situation.

Also the choice of α and β is special in this example. As Emily highlights, the annihilating operator being associated to the class of the whole space, $[X]$, allows us to remove any point $y \in X$. If we chose another class β and the support of the quotient $\mathcal{J}'/\mathcal{J}$ contained only points that are not in β, the operator would kill the class. This means that if we iterate an annihilating operator enough times we arrive at the zero class. Does this have any interpretation (maybe in vertex algebras)?

8. Proof of the main theorem

We split the proof in two different cases:

(1) $i, j > 0$ or $i, j < 0$, and
(2) $i > 0$, $j < 0$ or $i < 0$, $j > 0$.

Only in case (2) can the commutator be nonzero.

Case (1). Suppose $i, j > 0$. The intuitive picture in this case is that removing i points and then removing j more points would coincide with removing j and then removing i. More precisely: starting with any set of n points, the composition $P_\alpha[i]P_\beta[j]$ acts by removing j points so that the support of the quotient is in the class β, and then remove i more points so that the support of the quotient is in α. Then we would check that this operation yields the same class if we do it in the opposite order.

To do this formally we have to use our general definition for composition of correspondences, Equation 5.2.1. In view of the particular correspondences we are using, Equation 6.0.1, we see that the composition $P_\alpha[i]P_\beta[j]$ is given by

$$P_\alpha[i]P_\beta[j] = p_{13,*}(p_{12}^*[P[i]] \cap \Pi_1^*\alpha \cap p_{23}^*[P[j]] \cap \Pi_2^*\beta). \tag{8.0.1}$$

Forgetting about the classes α and β for a minute, consider the set

$$\begin{aligned} P[i,j] &:= p_{12}^{-1}(P[i]) \cap p_{23}^{-1}(P[j]) \cap (X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]}) \\ &= \left\{ (\mathcal{J}_1, \mathcal{J}_2, \mathcal{J}_3) : \mathcal{J}_1 \supset \mathcal{J}_2 \supset \mathcal{J}_3, \substack{\mathrm{supp}(\mathcal{J}_1/\mathcal{J}_2)=\{x\} \text{ for some } x\in X \\ \mathrm{supp}(\mathcal{J}_2/\mathcal{J}_3)=\{y\} \text{ for some } y\in X} \right\} \end{aligned} \tag{8.0.2}$$

which corresponds to the diagram

$$\begin{tikzcd} & X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]} \arrow[dl, "p_{12}"'] \arrow[dr, "p_{23}"] \arrow[dd, "p_{13}"] & \\ P[j] \subset X^{[n-j]} \times X^{[n]} & & X^{[n-i-j]} \times X^{[n-j]} \supset P[i] \\ & X^{[n-i-j]} \times X^{[n]} & \end{tikzcd}$$

This is to be read as follows. First keep in mind that a correspondence may be equivalently thought of as a subvariety, a homology class and an operator in homology. By definition, the correspondence $P[j]$ maps the homology of $X^{[n]}$ to the homology of $X^{[n-j]}$. In order to compose this with $P[i]$ we must regard $P[i]$ as a map from the homology of $X^{[n-j]}$ to the homology of $X^{[n-j-i]}$, which is perfectly valid since our correspondences are defined for all n.

Observe that the middle factor $X^{[n-j]}$ in the triple product of will become $X^{[n-i]}$ when we compute the composition in the opposite order. Being subsets of different products, it not immediately clear that the associated operators will coincide.

To retrieve the correspondence as in Equation 8.0.1 from the set $P[i,j]$ we shall consider its algebraic closure and to it associate a fundamental class. However, we will do a key procedure before that. Observe that $P[i,j]$ may be decomposed as disjoint union of the set $\{x \neq y\}$ of "generic" cases in which x and y in Equation 8.0.2 are different, and the set $\{x = y\}$ in which they coincide.

Let $\overline{P}[i,j] := \overline{P[i,j] \cap \{x \neq y\}}$. Then

$$p_{12}^*[P[i]] \cap p_{23}^*[P[j]] = [\overline{P}[i,j]] + \iota_* R$$

where $R \in H_*^{BM}(P[i,j] \cap \{x = y\})$ is the part corresponding to the "singular" part of $P[i,j]$. Here ι is the inclusion of $\{i = j\}$ into $P[i,j]$.

To arrive at Equation 8.0.1 the next step is to intersect with the classes $\Pi_1^*\alpha$ and $\Pi_2^*\beta$. Observe these commute with the classes $P_{12}^*[P[i]]$ and $[P_{23}^*[P[j]]$ since they correspond to disjoint components of the product $X^{[n-j]} \times X^{[n]} \times X$. Then it only remains to pushforward by p_{13}.

We can carry out the same construction for the composition $P_\beta[j]P_\alpha[i]$, obtaining the expressions

$$P_\alpha[j]P_\beta[i] = p_{13,*}(p_{12}^*[P[j]] \cap \Pi_2^*\alpha \cap p_{23}^*[P[i]] \cap \Pi_1^*\beta). \tag{8.0.3}$$

and

$$p_{12}^*[P[j]] \cap p_{23}^*[P[i]] = [\overline{P}[j,i]] + \iota'_* R'. \tag{8.0.4}$$

Observe that Equations 8.0.1 and 8.0.3 differ in the classes $\Pi_1^*\alpha$ against $\Pi_2^*\alpha$, and correspondingly with β. It is clear that the following two lemmas are all we need to confirm that the difference of these two equations vanishes (which is the commutation relation we are aiming for):

Lemma 8.1. (1) *There exists an isomorphism*

$$P[i,j] \cap \{x \neq y\} \xrightarrow{\cong} P[j,i] \cap \{x \neq y\}, \tag{8.1.1}$$

which exchanges Π_1 *and* Π_2.

(2) *We have*

$$p_{13,*}(\Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R) = 0, \qquad p_{13,*}(\Pi_1\beta \cap \Pi_2^*\alpha \cap \iota_* R') = 0. \tag{8.1.2}$$

Proof. (1) First notice that when we say the isomorphism, say φ, "exchanges Π_1 and Π_2" we mean that $\Pi_1|_{X\times X} = \Pi_2|_{X\times X} \circ \varphi$ according to the diagram

$$\begin{tikzcd}
 & & X \\
P[i,j] \cap \{x \neq y\} \subset \arrow[ddd, "\varphi"'] & X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]} \times X \times X \arrow[ur, "\Pi_1"] \arrow[dr, "\Pi_2"'] & \\
 & & X \\
 & & X \\
P[j,i] \cap \{x \neq y\} \subset & X^{[n-i-j]} \times X^{[n-i]} \times X^{[n]} \times X \times X \arrow[ur, "\Pi_1"] \arrow[dr, "\Pi_2"'] & \\
 & & X.
\end{tikzcd}$$

Now, the point of the map φ is that it must exchange an ideal in $X^{[n-j]}$ for an ideal in $X^{[n-i]}$. More precisely, we map $(\mathcal{J}_1, \mathcal{J}_2, \mathcal{J}_3) \in P[i,j] \cap \{x \neq y\}$ to $(\mathcal{J}_1, \mathcal{J}_2', \mathcal{J}_3) \in P[j,i] \cap \{x \neq y\}$. The defining relation is

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_2/\mathcal{J}_3, \qquad \mathcal{J}_2'/\mathcal{J}_3 = \mathcal{J}_1/\mathcal{J}_2. \tag{8.1.3}$$

No further explanation other than [Nak, Firgure 8.2] is provided.

The setting is the following. We have the ideals with contentions $\mathcal{J}_1 \supset \mathcal{J}_2, \mathcal{J}_2' \supset \mathcal{J}_3$. Passing to the supports, the inclusions are reversed, i.e the support of $\mathcal{J}_3$ contains the rest, and the support of $\mathcal{J}_1$ is contained in the rest.

To prove that this function is well-defined we need to show that for every $\mathcal{J}_2$ there is a unique $\mathcal{J}_2'$ satisfying Equation 8.1.3. Notice that by definition, $\mathrm{supp}\mathcal{J}_2/\mathcal{J}_3 = \{x\}$ for some $x \in X$ so that, as a scheme, $\{x\}$ has multiplicity j. We must define $\mathcal{J}_2' = \mathcal{J}_1 \cup \{x\}$ where x has multiplicity j. (Caveat: how can we make sure that indeed there is only one scheme of length j supported in the point x? Recall that elements in the Hilbert scheme should not be considered barely as sets of points with multiplicities! Actually, in Proposition 6.1 we used the fact that the space of possible choices coincides $\pi^{-1}(S^j_{(j)})$ and has dimension $j-1$.)

Defining $\mathcal{J}_2'$ this way implies that (recall we agreed about being sloppy with our notations of the supports and multiplicities)

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_1/(\mathcal{J}_1 \cup \{x\}) = \{x\} = \mathcal{J}_2/\mathcal{J}_3.$$

Now we also must have $\mathcal{J}_1/\mathcal{J}_2 = \{y\}$ for some $y \in X$ with multiplicity i. This implies that $\mathcal{J}_1/\mathcal{J}_3 = \{x\} \cup \{y\}$ with their multiplicities. Then

$$\mathcal{J}_2'/\mathcal{J}_3 = (\mathcal{J}_1 \cup \{x\})/\mathcal{J}_3 = \{y\} = \mathcal{J}_1/\mathcal{J}_2$$

as required. An inverse function defined similarly shows this is a bijection.

Here I am supposing that "isomorphism" means bijection since $P[i,j]$ is considered as a set-theoretical intersection.

(2) (Outline.) First we argue that

$$\begin{aligned}\Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R &= \dots \\ &= \iota_*(\Pi'^*(\alpha \cap \beta) \cap R.\end{aligned}$$

Then we argue that

$$p_{13,*}\iota_*(\Pi'^*(\alpha \cap \beta) \cap R) = \Pi''^*(\alpha \cap \beta) \cap p_{13,*}\iota_*(R). \tag{8.1.4}$$

By dimension reasons (not clear to me), $\iota_*(R)$ must vanish.

□

9. Götsche's formula

So far I didn't investigate how this formula can be derived from Theorem 6.2. This is not explained in the videos but most likely it is in [Nak].

For any variety Z (or manifold) we can form the *Poincaré polynomial* of Z, which is the generating function of the Betti numbers of Z

$$P_t(Z) = \sum_{m \geq 0} b_m(Z) t^m, \qquad b_m(Z) = \mathrm{rank} H_m(Z).$$

The idea is to consider the Poincaré polynomials for each each of the n-th Hilbert schemes and look at a generating function of all these polynomials That is, this is some sort of "bi-generating function in t and n":

$$\sum_{n \geq 0} q^n P_t(X^{[n]}) = \text{[Nak95, Equation 1.1]}.$$

References

[Har77] Robin Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, vol. 52, Springer, 1977.

[Hat00] A. Hatcher, *Algebraic topology*, Cambridge Univ. Press, Cambridge, 2000.

[HM98] Joe Harris and Ian Morrison, *Moduli of curves*, Graduate Texts in Mathematics, vol. 187, Springer-Verlag, 1998.

[Kac10] Victor Kac, *Lecture notes of 18.745 – introduction to lie algebras (fall 2010)*, `https://math.mit.edu/classes/18.745/classnotes.html`, 2010, Lecture notes, MIT.

[Kam10] Joel Kamnitzer, *Lectures on geometric constructions of the irreducible representations of gl_n*, 2010.

[Nak] Hiraku Nakajima, *Lectures on hilbert schemes of points on surfaces*, University lecture series, American Mathematical Soc.

[Nak95] ———, *Heisenberg algebra and hilbert schemes of points on projective surfaces*, 1995.

[vE23] Jethro van Ekeren, *Lecture notes of vertex algebras (verão 2023)*, `https://w3.impa.br/~jethro/2023-0/notes-2023-02-28`, 2023, Lecture notes, IMPA.